

IRIS FLOWER CLASSIFICATION USING MACHINE LEARNING

1. Introduction

Iris Flower Classification is a supervised machine learning project that predicts the species of an Iris flower using its measurements. The model classifies flowers into Setosa, Versicolor, and Virginica species.

2. Objective

To build a machine learning model capable of accurately predicting Iris flower species.

3. Dataset Description

Dataset Source: Kaggle Iris CSV Dataset. The dataset contains 150 flower records with Sepal Length, Sepal Width, Petal Length, Petal Width, and Species.

4. Tools and Technologies

Python, Pandas, Scikit-Learn, Jupyter Notebook

5. Algorithm Used

K-Nearest Neighbors (KNN)

6. Methodology

1. Data Collection
2. Data Preprocessing
3. Train-Test Split
4. Model Training
5. Prediction
6. Performance Evaluation

7. Results

The KNN model successfully classified Iris flower species with high accuracy.

8. Conclusion

This project demonstrates the application of supervised machine learning for classification tasks. The KNN algorithm effectively predicts Iris flower species based on flower measurements.